

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 29-2023: A 34-Year-Old Woman with Abdominal Distention and Acute Kidney Injury

Sundar Jagannath, M.B., B.S., Mark A. Anderson, M.D., Andrew J. Yee, M.D.,
and Robert P. Hasserjian, M.D.

PRESENTATION OF CASE

Dr. Matthew B. Roberts (Medicine): A 34-year-old woman with a history of end-stage liver disease due to alcohol use disorder was admitted to this hospital because of abdominal distention, fatigue, and acute kidney injury that occurred 10 months after orthotopic liver transplantation.

Two weeks before the current admission, fatigue, weakness, and anorexia developed, along with associated myalgias and night sweats. Three days before the current admission, three to five episodes of nonbloody, watery diarrhea occurred.

The next day, symptoms did not abate, and the patient presented to another hospital for evaluation. She reported no fever, chills, nausea, vomiting, abdominal pain, dyspnea, or swelling of the legs. The temporal temperature was 36.6°C, the blood pressure 76/49 mm Hg, the pulse 132 beats per minute, the respiratory rate 20 breaths per minute, and the oxygen saturation 100% while she was breathing ambient air. She appeared thin and frail. There was abdominal distention without tenderness. The remainder of the examination was unremarkable. The blood creatinine level was 3.25 mg per deciliter (287 μ mol per liter; reference range, 0.55 to 1.30 mg per deciliter [49 to 115 μ mol per liter]), which was higher than the level obtained 1 month earlier (1.27 mg per deciliter [112 μ mol per liter]). The blood sodium level was 120 mmol per liter (reference range, 136 to 146) and the potassium level 6.3 mmol per liter (reference range, 3.6 to 5.2); additional laboratory test results are shown in Table 1.

Three liters of fluids, hydrocortisone, piperacillin–tazobactam, and vancomycin were administered intravenously, and the blood pressure increased. Computed tomography (CT) of the abdomen, performed without the administration of intravenous contrast material, revealed a large volume of ascitic fluid and scattered areas of high-density material. Paracentesis was performed, and 1300 ml of ascitic fluid was removed. Analysis of the ascitic fluid was notable for more than 1000 red cells per microliter, 2099 nucleated cells per microliter (with 400 neutrophils per microliter), and numerous atypical cells that were not further characterized. Blood and ascitic fluid specimens were obtained for culture.

From the Department of Hematology–Oncology, Mount Sinai Hospital, and the Department of Hematology–Oncology, Icahn School of Medicine at Mount Sinai — both in New York (S.J.); and the Departments of Radiology (M.A.A.), Medicine (A.J.Y.), and Pathology (R.P.H.), Massachusetts General Hospital, and the Departments of Radiology (M.A.A.), Medicine (A.J.Y.), and Pathology (R.P.H.), Harvard Medical School — both in Boston.

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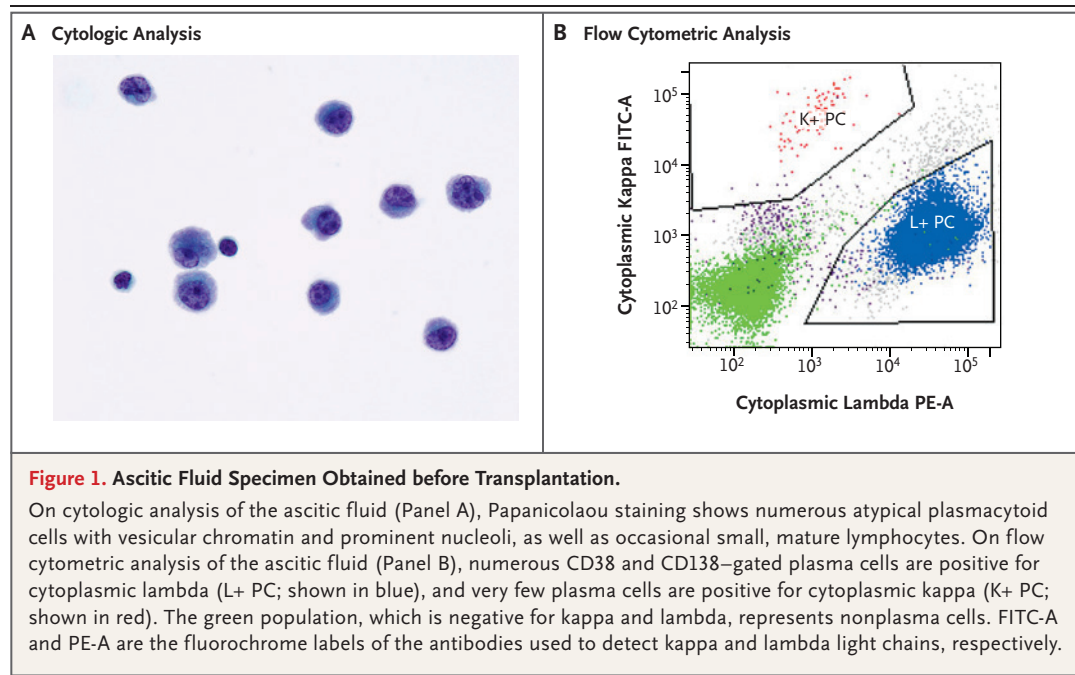
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Table 1. Laboratory Data.*

Variable	Reference Range, Other Hospital	1 Day before Current Admission, Other Hospital	Reference Range, Adults, This Hospital†	On Current Admission, This Hospital	Day 3, This Hospital
Blood					
Hemoglobin (g/dl)	11.8–16.0	9.0	12.0–16.0	7.2	8.6
Hematocrit (%)	36.0–47.0	25.7	36.0–46.0	21.8	25.6
Platelet count (per μ l)	150,000–400,000	308,000	150,000–400,000	176,000	171,000
White-cell count (per μ l)	3700–11,200	13,500	4500–11,000	6290	7730
White-cell differential count (per μ l)					
Neutrophils	1480–7950	10,560	1800–7700	5390	5270
Lymphocytes	740–5040	1940	1000–4800	480	1650
Monocytes	0–1340	870	20–1200	350	660
Eosinophils	0–450	0	0–90	0	30
Basophils	0–220	20	0–30	10	30
Sodium (mmol/liter)	136–146	120	135–145	129	133
Potassium (mmol/liter)	3.6–5.2	6.3	3.4–5.0	4.5	4.2
Chloride (mmol/liter)	98–110	87	98–108	93	—
Carbon dioxide (mmol/liter)	21–32	21	23–32	20	16
Anion gap (mmol/liter)	3–11	12	3–17	16	20
Glucose (mg/dl)	70–110	75	70–110	127	—
Calcium (mg/dl)	8.5–10.5	9.5	8.5–10.5	8.0	8.3
Phosphorus (mg/dl)	—	—	2.6–4.5	3.2	2.6
Urea nitrogen (mg/dl)	6–20	74	8–25	65	46
Creatinine (mg/dl)	0.55–1.30	3.25	0.60–1.50	3.08	2.25
Lactic acid (mmol/liter)	0.4–2.0	0.7	0.5–2.0	0.8	7.8
Alanine aminotransferase (U/liter)	6–55	24	7–33	17	20
Aspartate aminotransferase (U/liter)	6–40	52	9–32	39	42
Alkaline phosphatase (U/liter)	30–117	100	30–100	91	117
Total bilirubin (mg/dl)	0.2–1.2	0.5	0.0–1.0	0.3	0.4
Total protein (g/dl)	6.0–8.4	8.9	6.0–8.3	6.9	7.2
Albumin (g/dl)	3.2–5.0	1.9	3.3–5.0	1.9	2.3
Globulin (g/dl)	1.9–4.1	7.0	1.9–4.1	5.0	4.9
Lactate dehydrogenase (U/liter)	—	—	110–210	—	697
Uric acid (mg/dl)	—	—	2.3–6.6	—	21.2
Ascitic fluid					
Nucleated-cell count (per μ l)	0–305	2099	—	—	2018
Nucleated-cell differential count (%)					
Neutrophils	—	19	—	—	11
Macrophages	—	0	—	—	28
Lymphocytes	—	2	—	—	28
Monocytes	—	11	—	—	3
Unclassified	—	68	—	—	30
Red-cell count (per μ l)	0–24,999	<1000	—	—	16,000

* To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for phosphorus to millimoles per liter, multiply by 0.3229. To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for lactic acid to milligrams per deciliter, divide by 0.1110. To convert the values for bilirubin to micromoles per liter, multiply by 17.1. To convert the values for uric acid to micromoles per liter, multiply by 59.48.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.



On the second hospital day, the patient was transferred to the intensive care unit (ICU) of this hospital because of concern for sepsis. She reported persistent fatigue, weakness, myalgias, and watery diarrhea.

Additional history was obtained. Twenty-two months before this admission, the patient had been admitted to the other hospital because of abnormal results on liver-function tests. She had received a diagnosis of alcoholic hepatitis complicated by cirrhosis with ascites. Treatment with a short course of prednisone had been initiated, furosemide administered, and large-volume paracentesis performed for the management of ascites. The patient had been advised to discontinue alcohol consumption. Ten months later, and 12 months before the current admission, the patient had been admitted to this hospital because of decompensated liver disease complicated by hepatorenal syndrome. Renal replacement therapy had been initiated, and small-volume paracentesis had been performed to obtain ascitic fluid for cytologic and hematologic analysis.

Dr. Robert P. Hasserjian: Analysis of the ascitic fluid performed 12 months before the current admission had been notable for 44 nucleated cells per microliter, of which 1% were lymphocytes, 53% monocytes, 17% macrophages, and

29% atypical-appearing plasma cells (Fig. 1A). These findings had prompted flow cytometric analysis of the ascitic fluid, which had revealed a clonal plasma cell population with lambda light-chain restriction (Fig. 1B) but no evidence of a monoclonal B-cell or unusual T-cell population.

Dr. Roberts: Twelve months before the current admission, the serum IgG level had been 1394 mg per deciliter (reference range, 614 to 1295), the IgA level 885 mg per deciliter (reference range, 69 to 309), and the IgM level 177 mg per deciliter (reference range, 53 to 334). Serum protein electrophoresis had revealed a diffuse increase in the gamma globulin level in a normal pattern; no monoclonal protein had been detected on immunofixation. The serum level of free kappa light chains had been 246.2 mg per liter (reference range, 3.3 to 19.4) and the level of free lambda light chains 78.4 mg per liter (reference range, 5.7 to 26.3), with a kappa:lambda ratio of 3.14 (reference range, 0.26 to 1.65). A bone marrow biopsy had been performed, and analysis of the bone marrow had revealed 5% plasma cells, which were polytypic for kappa and lambda light chains on in situ hybridization. Flow cytometric analysis of the bone marrow had revealed no clonal plasma cells or abnormal lymphocytes. ¹⁸F-fluorodeoxyglucose (FDG) positron-emission

tomography (PET) and CT of the chest, abdomen, and pelvis had revealed a cirrhotic liver with evidence of portal hypertension, including abdominal varices and ascites, but no focal masses or lymphadenopathy.

Taken together, these findings had been considered to be due to liver disease, an inflammatory state, and kidney dysfunction, and the patient had been listed for liver transplantation. Ten months before the current admission, orthotopic liver transplantation had been performed. Pretransplantation testing for Epstein-Barr virus (EBV) IgG and cytomegalovirus (CMV) IgG had been positive in both the donor and the recipient. The explanted native liver had shown micronodular cirrhosis with mildly inflamed septa and evidence of iron overload. The immediate post-transplantation course had been notable for intraperitoneal hemorrhage, as well as vancomycin-resistant *Enterococcus faecium* bacteremia and recurrent peritonitis. The patient had been discharged to a rehabilitation facility 21 days after transplantation.

Three months after liver transplantation, and 7 months before the current admission, biliary stent obstruction and multidrug-resistant *Klebsiella pneumoniae* bacteremia had developed. The biliary stent had been removed and intravenous antibiotic agents administered. Four months before the current admission, asymptomatic CMV viremia had developed. Outpatient treatment with valganciclovir had been initiated. The patient had remained well until the current admission.

At the time of the current admission, medications included aspirin, atovaquone, vitamin B complex, calcium citrate, vitamin D₃, ferrous sulfate, fludrocortisone, gabapentin, magnesium, melatonin, mycophenolate, omeprazole, prednisone, tacrolimus, and ursodiol. There were no known drug allergies. The patient was born in New England and lived with her mother. She had not traveled outside the region, and there had been no animal exposures. The patient had stopped drinking alcohol and smoking cigarettes approximately 2 years before the current admission. There was no illicit drug use. Her mother and father both had cardiac and liver disease; her maternal grandfather also had liver disease.

On examination, the temperature was 36.7°C, the blood pressure 100/53 mm Hg, the heart rate

114 beats per minute, and the oxygen saturation 100% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 21. The patient appeared thin and frail. An abdominal examination revealed moderate distention, shifting dullness, and softness without tenderness. There was no hepatosplenomegaly or lymphadenopathy. The remainder of the examination was unremarkable. Laboratory test results are shown in Table 1. Treatment with intravenous cefepime and linezolid and oral vancomycin was initiated, and intravenous fluids were continued. Blood, urine, and stool specimens were obtained for culture.

Dr. Mark A. Anderson: PET-CT of the chest, abdomen, and pelvis revealed multifocal, predominantly peritoneal FDG-avid masses, as well as diffuse FDG uptake along the surface of large-bowel and small-bowel loops (Fig. 2). These findings had not been present on a CT scan obtained 1 year earlier.

Dr. Roberts: Testing for *Clostridioides difficile* toxin in stool was positive. Treatment with intravenous antibiotics was stopped, and oral vancomycin was continued. The diarrhea abated, and the hypotension resolved. On the third hospital day, the blood creatinine level decreased to 2.25 mg per deciliter (199 μ mol per liter); other laboratory test results are shown in Table 1. Blood, urine, and stool cultures were negative. Diagnostic paracentesis was performed, and analysis of the ascitic fluid was notable for 2018 nucleated cells per microliter, of which 30% were unclassified; additional results are shown in Table 1.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Sundar Jagannath: This 34-year-old woman had a history of alcoholic hepatitis that was complicated by cirrhosis with ascites. Hepatorenal syndrome developed, and she was treated with renal replacement therapy followed by liver transplantation. Her post-transplantation course was complicated by multiple infections, for which she received antibiotics. The current hospitalization, 10 months after liver transplantation, was precipitated by *C. difficile* infection that resulted in diarrhea, hypotension, tachycardia, acute kidney injury, and metabolic disarray. After the

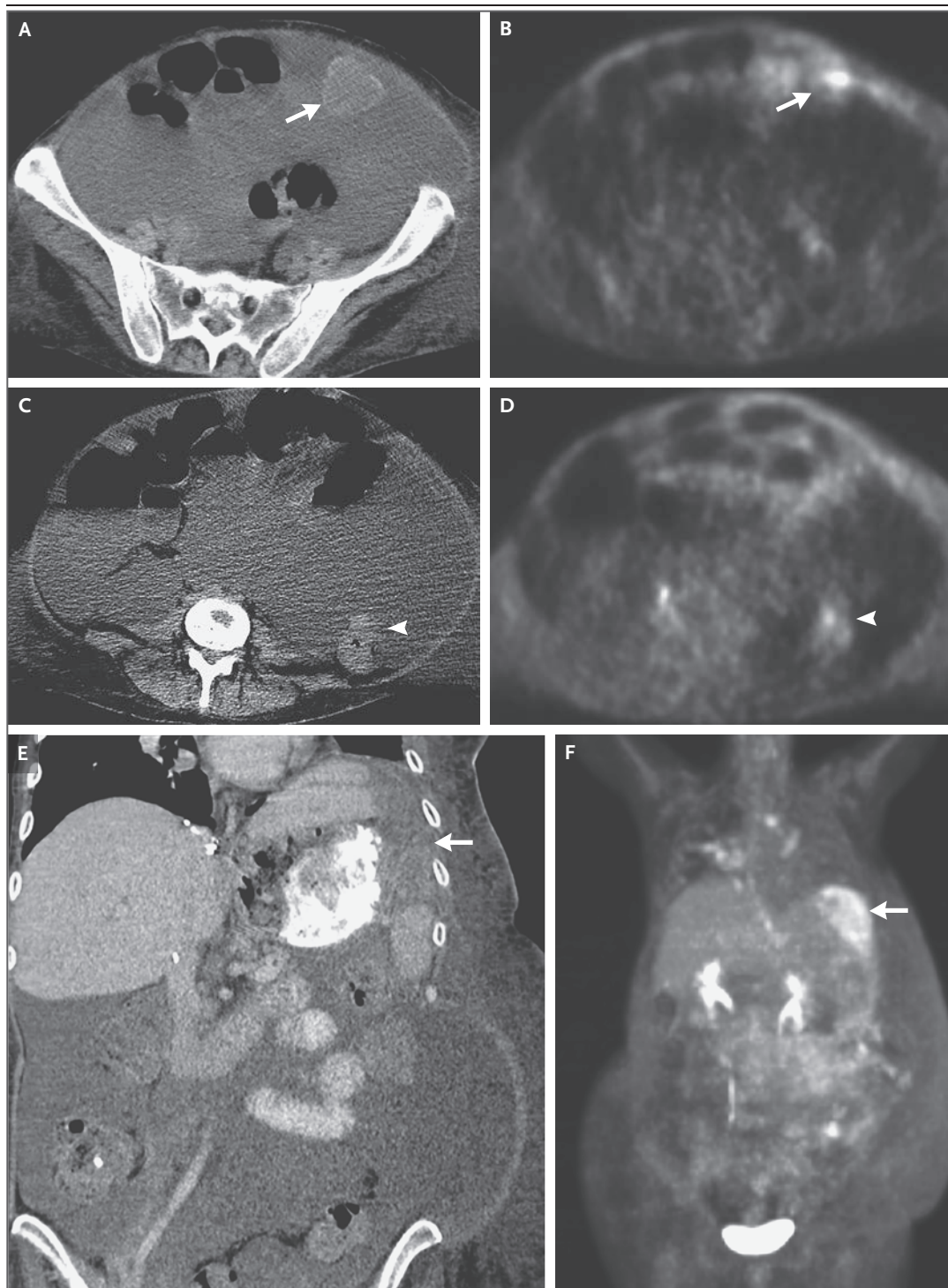


Figure 2. Abdominal Images.

CT with the administration of oral contrast material and ^{18}F -fluorodeoxyglucose (FDG) positron-emission tomography were performed. Axial images through the abdomen and pelvis (Panels A through D) and coronal reconstruction images (Panels E and F) show FDG-avid peritoneal masses in the left greater omentum and left upper quadrant (Panels A, B, E, and F; arrows) and diffuse serosal thickening with associated FDG uptake along the surface of the large bowel (Panels C and D; arrowheads). These findings were not present on a CT scan obtained 1 year earlier.

patient received intravenous fluids and oral vancomycin for the treatment of *C. difficile* infection, the diarrhea abated and the hypotension resolved. However, during the current illness, imaging of the abdomen and pelvis revealed large-volume ascites, as well as multifocal, predominantly peritoneal FDG-avid masses and diffuse, moderate FDG uptake along the surface of multiple bowel loops.

INFECTION

In constructing a differential diagnosis for these abnormal imaging findings, I will first consider infectious causes. This patient was immunosuppressed and had a history of recurrent peritonitis. Patients with tuberculous peritonitis can present with ascites. They can also present with loculated intraabdominal abscess, but this patient did not have evidence of matted bowel and omentum preventing the free flow of ascitic fluid on PET-CT. In addition, she did not have other PET-CT findings that are typical of tuberculous peritonitis, including enlarged, possibly necrotic retroperitoneal and mesenteric lymph nodes and thickened omentum. The absence of fever and constitutional symptoms makes tuberculous peritonitis unlikely, although immunosuppression could have blunted the development of fever.

CANCER

This patient's ascites is most likely related to a solid or hematologic cancer. She could have peritoneal carcinomatosis due to ovarian, gastric, or breast cancer. We have not been told whether there was an abnormal breast examination in this patient. Imaging did not reveal an ovarian or gastric mass, instead showing predominantly peritoneal masses. Patients with primary peritoneal mesothelioma can present with large peritoneal masses that may be nodular, with or without ascites. Although primary peritoneal mesothelioma is very rare, it remains a possibility in this case.

PRIMARY EFFUSION LYMPHOMA

Patients with primary effusion lymphoma, which is a large B-cell lymphoma, usually present with serous effusions (e.g., pleural effusion or ascites), without masses. Primary effusion lymphoma most often occurs in persons with human immunodeficiency virus (HIV) infection but can also occur in those who have undergone solid-

organ transplantation. Primary effusion lymphoma has been associated with human herpesvirus 8 (HHV-8) infection. We have not been specifically given the patient's status with respect to HHV-8 infection, but she has been followed in the transplantation clinic while receiving immunosuppression and would have been monitored for reactivation of EBV infection. Although I cannot rule out primary effusion lymphoma, I will consider other, more likely explanations for her presentation.

POST-TRANSPLANTATION LYMPHOPROLIFERATIVE DISORDER

Post-transplantation lymphoproliferative disorder (PTLD) that is associated with EBV commonly causes lymphadenopathy, hepatosplenomegaly, subcutaneous nodules, or tonsillar enlargement, and none of these features were present in this patient. Ascites can be a presenting sign of PTLD, although it is rare^{1,2}; therefore, PTLD remains a possible diagnosis in this case. The ascitic fluid analysis performed at the time of the current admission revealed 2018 nucleated cells per microliter, of which 11% were neutrophils and 30% unclassified. This finding leads me to consider the clinical significance of the atypical cells noted on the ascitic fluid analysis performed before transplantation. Flow cytometric analysis of the atypical cells that was performed at that time revealed a clonal plasma cell population with lambda light-chain restriction. This finding led to a serum workup, which showed hypergammaglobulinemia.

HYPERGAMMAGLOBULINEMIA

The differential diagnosis for hypergammaglobulinemia in this patient before liver transplantation includes benign conditions and cancers. Chronic inflammatory diseases (e.g., rheumatoid arthritis, scleroderma, or Hashimoto's thyroiditis), chronic skin diseases (e.g., pyoderma gangrenosum or necrobiotic xanthogranulomatosis), and chronic infections (e.g., tuberculosis, HIV infection, or bacterial endocarditis) are benign causes of hypergammaglobulinemia. This patient did not have any of these conditions, but she did have chronic liver disease, which is another benign cause of hypergammaglobulinemia. Cancers that can cause hypergammaglobulinemia (e.g., plasma cell disorders, lymphoma, or leukemia) should also be considered. Plasma cell

disorders include monoclonal gammopathy of undetermined significance (MGUS), asymptomatic (smoldering) multiple myeloma, multiple myeloma, and AL amyloidosis (also known as primary or light-chain amyloidosis).

Before transplantation, this patient was evaluated for plasma cell disorders. Serum protein electrophoresis and immunofixation revealed that the hypergammaglobulinemia was polyclonal, a finding confirmed by increased levels of IgG and IgA with a normal level of IgM. Levels of both free kappa and free lambda light chains were elevated, with a ratio of 3.14. This ratio is higher than the normal range (0.26 to 1.65) but within the reference range for a patient receiving hemodialysis (0.37 to 3.10), as this patient was at the time. Evaluation of the bone marrow was negative for a clonal plasma cell population. PET-CT showed a cirrhotic liver with evidence of portal hypertension, including upper abdominal varices and ascites, but no lymphadenopathy or masses.

Therefore, the patient's polyclonal hypergammaglobulinemia was attributed to her known liver disease. The liver plays an important role in removing antigens and endotoxins that arise from the gut. In patients with chronic liver disease that is complicated by portal hypertension, antigens and endotoxins from the gut bypass the liver through collateral circulation and reach the antibody-producing cells, resulting in hypergammaglobulinemia.³ In such patients, the gamma globulin level returns to normal within a week after liver transplantation.⁴

However, this patient presented 10 months after liver transplantation with a persistently elevated globulin gap, as well as new anemia and persistent kidney injury, features that suggest the possibility of multiple myeloma. To determine whether this patient had multiple myeloma, I would perform a cytologic evaluation of the ascitic fluid or of a biopsy specimen of the peritoneal mass.

DR. SUNDAR JAGANNATH'S
DIAGNOSIS

Multiple myeloma.

PATHOLOGICAL DISCUSSION

Dr. Hasserjian: The diagnostic procedure in this case was a core biopsy of the mesenteric mass.

The mass contained sheets of large cells (Fig. 3A) that were positive for MUM1, CD138, CD56, and lambda light chains (Fig. 3B, 3C, and 3D) and were negative for CD20, CD19, kappa light chains, HHV-8, and EBV (on Epstein-Barr encoding region in situ hybridization). The large cells also strongly expressed MYC and had a Ki-67 proliferation index of 80%. Similar cells were seen in the ascitic fluid. Flow cytometric analysis of the ascitic fluid revealed that they were clonal plasma cells that expressed monotypic lambda light chains.

A bone marrow biopsy was also performed. The bone marrow contained clusters of large, neoplastic plasma cells with prominent nucleoli (Fig. 3E). On analysis of a smear of bone marrow aspirate, 15% of the cells were highly atypical plasma cells with prominent nucleoli (Fig. 3F). On flow cytometric analysis, these cells expressed monotypic lambda light chains. Fluorescence in situ hybridization (FISH) of the bone marrow aspirate revealed a MYC rearrangement, an IGH rearrangement, a 1q22 duplication, and a deletion of the TP53 locus at 17p13.1. Next-generation sequencing revealed pathogenic variants in HRAS, STAT3, and ARID1A. Serum protein electrophoresis and immunofixation showed an IgA lambda M component (2.28 g per deciliter).

The overall findings confirmed a diagnosis of plasma cell myeloma with plasmablastic features. Plasmablastic plasma cell myeloma has morphologic and immunophenotypic features similar to those of plasmablastic lymphoma, but plasmablastic lymphoma is usually positive for EBV and is not typically associated with a high serum paraprotein level. Although a clonal plasma cell population had been identified in the ascitic fluid in this patient before liver transplantation, the unusual fluid-based presentation in the absence of tissue masses, bone marrow involvement, and paraprotein had precluded a definitive diagnosis of plasma cell myeloma at that time; in rare cases, clonal plasma cells may be identified in fluid specimens in reactive conditions.^{5,6} Nevertheless, because this clonal plasma cell population had been present before the patient's liver transplantation, the plasma cell myeloma was not considered to be a PTLT.

PATHOLOGICAL DIAGNOSIS

Plasma cell myeloma (multiple myeloma) with plasmablastic features.

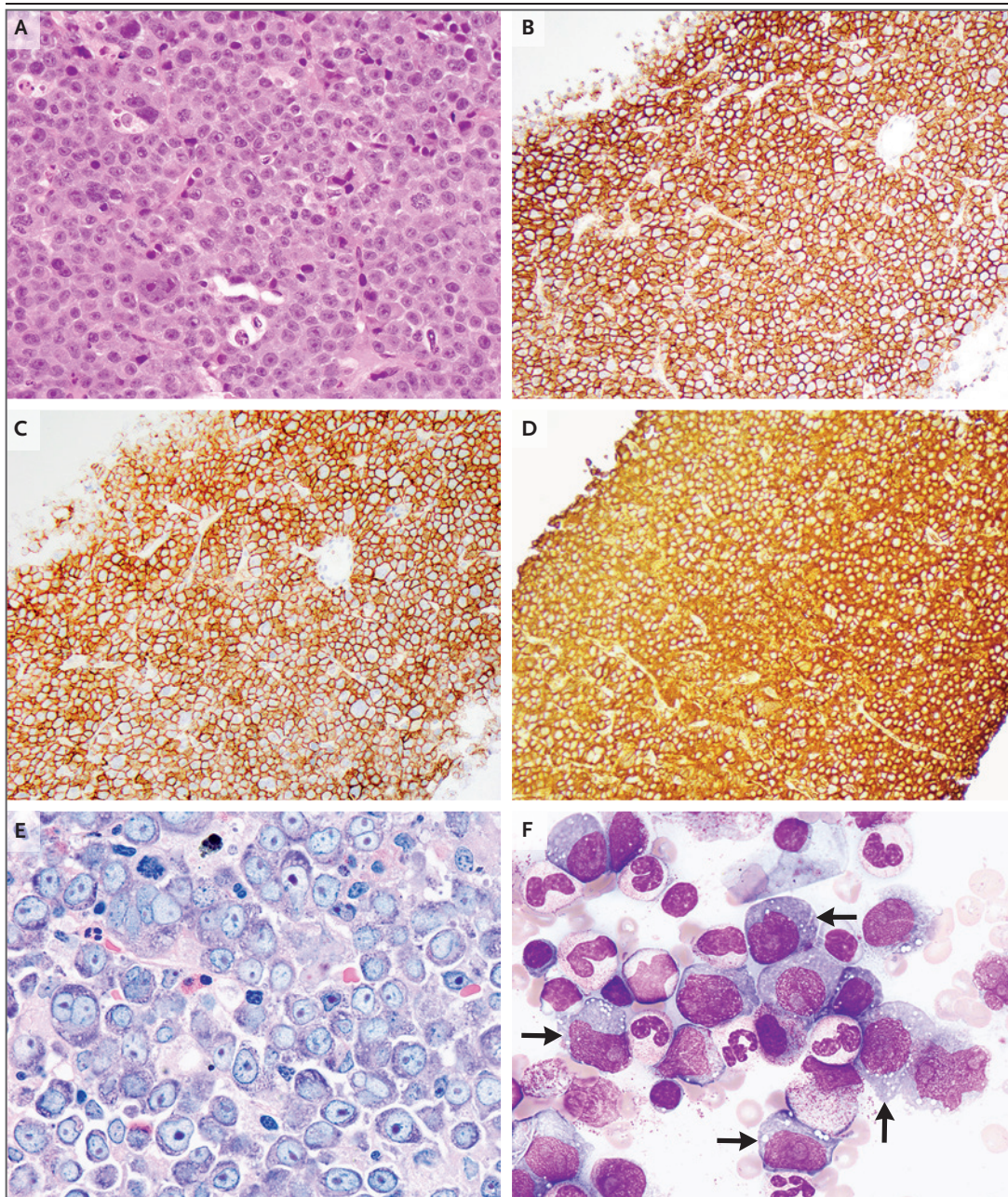


Figure 3. Biopsy Specimens of the Mesenteric Mass and Bone Marrow.

On hematoxylin and eosin staining, the mesenteric mass contains sheets of large pleomorphic cells with vesicular chromatin, prominent central nucleoli, and abundant, often eccentric cytoplasm (Panel A). On immunohistochemical staining, these large cells are positive for CD138 (Panel B), CD56 (Panel C), and lambda light chains (Panel D). On Giemsa staining, the bone marrow contains clusters of large cells with prominent central nucleoli and basophilic cytoplasm, features consistent with plasmablastic plasma cells (Panel E). On Wright–Giemsa staining, a smear of bone marrow aspirate shows neoplastic plasma cells with prominent nucleoli (Panel F, arrows).

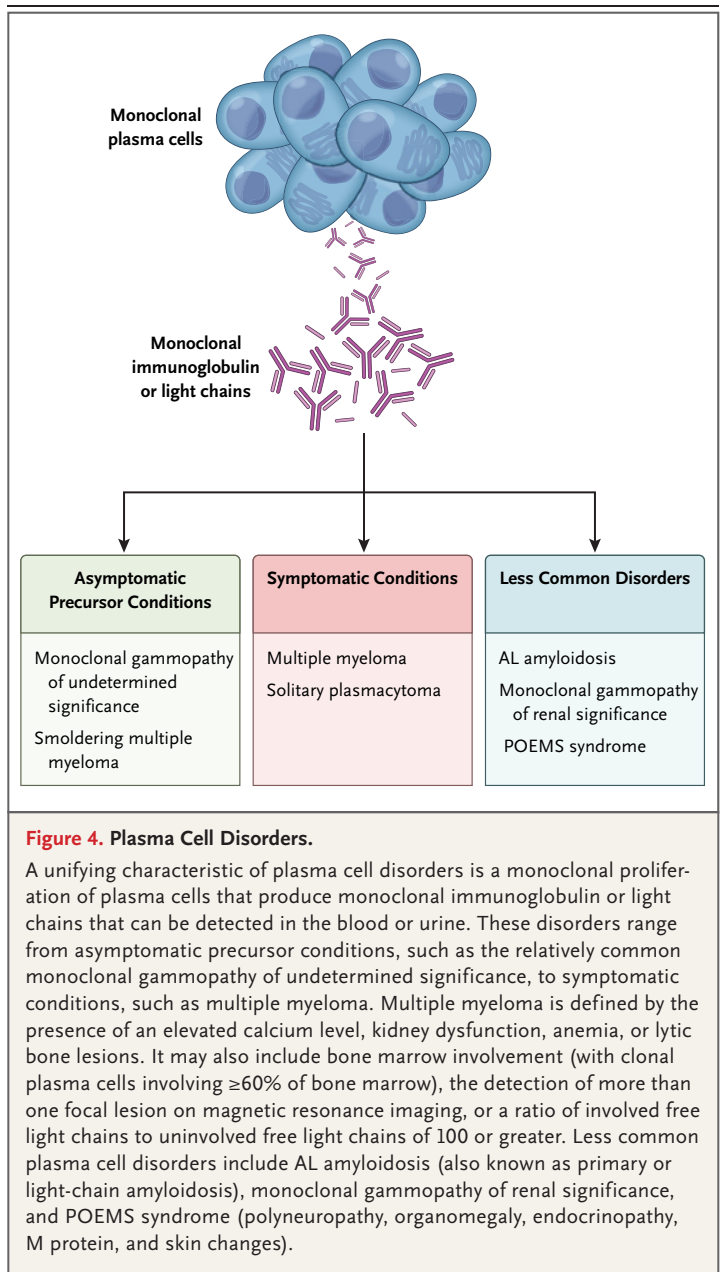
DISCUSSION OF MANAGEMENT

Dr. Andrew J. Yee: This patient had highly unusual manifestations of multiple myeloma. It can be helpful to consider multiple myeloma in the context of plasma cell disorders in general (Fig. 4).

A defining feature of plasma cell disorders is a clonal proliferation of plasma cells that produce monoclonal immunoglobulin or light chains that can be detected in the blood or urine. Plasma cell disorders range from asymptomatic conditions, such as MGUS (which is present in >3% of persons >50 years of age)⁷ and smoldering multiple myeloma, to symptomatic conditions, such as multiple myeloma and solitary plasmacytoma. Less common plasma cell disorders include AL amyloidosis and even rarer entities, such as monoclonal gammopathy of renal significance⁸ and POEMS syndrome (polyneuropathy, organomegaly, endocrinopathy, M protein, and skin changes).⁹ The mechanisms that allow a quiescent condition such as MGUS to transform into a life-altering condition such as multiple myeloma, which is characterized by acute kidney failure and pathologic fractures, are under active investigation.^{10,11}

Criteria for establishing the diagnosis of multiple myeloma are unique in that either pathological or clinical findings indicative of end-organ involvement are required. Multiple myeloma was traditionally defined by the presence of one or more of the four “CRAB” criteria: hypercalcemia, renal impairment, anemia, and bone involvement.¹² In 2014, the criteria were updated to include myeloma-defining biomarkers that could identify patients who are at high risk for the development of traditionally symptomatic disease: a ratio of involved free light chains to uninvolved free light chains of 100 or greater, bone marrow involvement (with clonal plasma cells involving $\geq 60\%$ of bone marrow), or the detection of more than one focal lesion on magnetic resonance imaging.

Unique features of this patient’s case include her young age, bulky extramedullary disease involving the peritoneum, and tumor lysis syn-



drome, along with her history of liver transplantation 10 months before diagnosis. The median age among persons with a new diagnosis of multiple myeloma is 69 years, and only 3.2% of cases of multiple myeloma are diagnosed in persons younger than 45 years of age.¹³ Extramedullary disease, which is disease outside the bone

marrow, is present in only 6% of patients with a new diagnosis.¹⁴ Spontaneous tumor lysis syndrome is uncommon in patients with multiple myeloma.

On presentation, this patient had an acute illness due to tumor lysis syndrome, which was probably related to the aggressive features of her disease. The first management priority was stabilization in the ICU. She received rasburicase, and her metabolic derangements were managed with continuous venovenous hemofiltration. Paracentesis was performed to relieve the symptoms of ascites. Once the diagnosis of multiple myeloma was made, treatment was initiated.

Treatment for multiple myeloma has evolved considerably in the past two decades. Before the 2000s, the main strategy for achieving a deep response was the administration of high-dose melphalan and autologous stem-cell transplantation. The introduction of several pivotal drug classes — proteasome inhibitors (e.g., bortezomib, carfilzomib, and ixazomib), immunomodulatory drugs (e.g., thalidomide, lenalidomide, and pomalidomide), and anti-CD38 monoclonal antibodies (daratumumab and isatuximab) — has transformed the treatment options, leading to improvements in quality of life and overall survival.¹⁵ These drugs are effective when administered in combinations, and they can be used in patients who are older and frailer.¹⁶ Moreover, treatments that target B-cell maturation antigens through chimeric antigen receptor T cells (idecabtagene vicleucel and ciltacabtagene autoleucel)^{17,18} or bispecific antibodies (teclistamab)¹⁹ have recently been approved and are likely to be helpful in the management of myeloma.

Patients with multiple myeloma typically receive outpatient treatment. A three-drug regimen — lenalidomide, bortezomib, and dexamethasone — is commonly used in the United States. This approach is based on the results of the SWOG S0777 trial,²⁰ which showed the superiority of this three-drug regimen over lenalidomide in combination with dexamethasone. The anti-CD38 monoclonal antibody daratumumab is increasingly being added to this three-drug regimen for the treatment of patients with newly diagnosed multiple myeloma who are eligible for intensive treatment, on the basis of the results of the GRIFFIN trial²¹ and the CASSIOPEIA trial.²² Urgent inpatient treatment may be considered for patients with acute

kidney injury due to cast nephropathy, in whom rapid reduction of light chains is important for the restoration of kidney function.

In this patient, we opted to initiate inpatient treatment, given the burden of symptomatic disease and associated tumor lysis syndrome. We selected a regimen of cyclophosphamide, bortezomib, and dexamethasone, which is commonly used in patients with newly diagnosed multiple myeloma and is also generally accessible in the inpatient setting.¹⁵ Although lenalidomide is a standard drug for the treatment of multiple myeloma, this drug can be associated with graft rejection because of its immunomodulatory effects, so it was avoided in this patient, given her history of liver transplantation.^{23,24}

Shortly after treatment was started, the patient's condition improved considerably. Continuous venovenous hemofiltration was discontinued. There was a rapid decrease in abdominal distention and in the size of the peritoneal masses. Monoclonal protein was no longer detected. The patient was discharged home 3 weeks after the initiation of treatment.

However, 2 months after the initiation of treatment, recurrent disease developed, with increasing abdominal masses and accumulation of ascites. This early relapse is probably a reflection of the underlying high-risk features of her disease, including the 17p deletion (which was present in 96% of cells) and the 1q duplication detected on FISH of the bone marrow aspirate.^{25,26} The patient had a prolonged hospitalization, during which she received a course of salvage cytoreductive chemotherapy with an infusion of dexamethasone, cyclophosphamide, and etoposide. She was ultimately discharged home, and her treatment was later changed to daratumumab administered as a single agent.^{27,28} Her condition improved initially, but 3 months later, she had rapid disease progression and died.

A physician: Could further testing before liver transplantation have led to the discovery of myeloma, and should the patient have undergone liver transplantation?

Dr. Jagannath: The combined DNA and cytoplasmic immunoglobulin assay for DNA ploidy by means of flow cytometry could have identified whether the plasma cells with light-chain restriction were aneuploid. FISH or array-based comparative genomic hybridization of the clonal plasma cells could have led to additional in-

sights. Finally, it could have been helpful to obtain the levels of free light chains in the ascitic fluid. If there were intraabdominal clonal plasma cells that were secreting excess light chains, the free light-chain levels in the ascitic fluid would have been much higher than the levels in the serum.²⁹

Dr. Yee: At the time that this patient was evaluated for liver transplantation, she was extremely ill, and it was likely that she would die within days to weeks without transplantation. We were concerned about the possibility of an undiagnosed plasma cell disorder, which could

have been an incidental finding. However, there was no clear evidence to support a diagnosis of multiple myeloma, so we made the decision to proceed urgently with lifesaving transplantation and address the possibility of myeloma at a later time.

FINAL DIAGNOSIS

Plasma cell myeloma (multiple myeloma).

This case was presented at Cancer Center Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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